Subject: Prospective MS/PhD Student (Fall 2027)

Dear Dr. Jeng,

I am writing to express my strong interest in joining the UI@NYCU lab as an MS or PhD student starting in Fall 2027. I recently came across the abstract for your 2024 IEEE APCCAS paper on automated design space optimization for FPGA-based sonar systems. Your work on automating hardware trade-offs to reduce latency and power consumption closely aligns with the problems I am currently working on in ultrasonic imaging. (As I no longer have institutional library access after graduating, I would love to read a preprint or author's copy if you happen to have one available.)

I graduated in Computer Science and Engineering from IIT Gandhinagar (9.59/10 CGPA) and currently work as a Data Engineer at Texas Instruments, where I build AI-assisted tools for high-level hardware block-diagram design. Alongside my full-time role, I am conducting independent research on hardware-efficient ultrasound beamforming. Specifically, I am developing a Cooperative Coevolutionary Genetic Algorithm (CCGA) to jointly optimize sparse transducer array layouts and non-linear activation functions for the Total Focusing Method (TFM). Because applying traditional p^{th}-root and p^{th}-power operations to partial images requires long computational paths on an FPGA, I am replacing them with learned activation functions, derived via symbolic and linear regression over hardware-friendly basis functions, that can be implemented with minimal LUT and DSP overhead.

Previously, at the MUSE Lab at IIT Gandhinagar, I designed and synthesized an end-to-end TFM accelerator on a Xilinx AC701 FPGA that achieves 30 fps reconstruction for a 128×128 grid. That project introduced a novel transducer domain partitioning scheme (accepted for a **poster presentation at IEEE SAUS 2026**) and was handed over to Modsonic Instruments for commercial NDT deployment.

Having worked across both academic FPGA research and commercial engineering, I am very interested in graduate research that connects directly to industry needs. NYCU's proximity to the Hsinchu Science Park and ITRI is a major draw for me, and I would be keen to contribute to industry-sponsored or co-developed projects within your lab.

I have attached my CV for your review. If you are taking students for Fall 2027, would you be open to a brief 15-minute call in the coming weeks to discuss how my background might fit with your upcoming projects?

Thank you for your time and consideration.

Best regards,

Pranav Joshi
Data Engineer, Texas Instruments

B.Tech CSE, IIT Gandhinagar
GitHub | LinkedIn | Academic Portfolio